

Cognitive Fatigue in Autoregressive Transformers

Independent Trackio logbook reproduction of paper #26805 / OpenReview dE3Z3bfEzk

2 verified
4 falsified
1 mixed

1. FI Formula Reproduces

The weighted index is well defined: 0.40 attention decay, 0.35 entropy penalty, 0.25 drift penalty.

150 GRID CHECKS	1.0 WEIGHT SUM	0.300 WORKED FI
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Boundedness and monotonicity held across all audited points.

2. Predictive Validity Falls Short

OPT-2.7B was evaluated on real HotpotQA, TriviaQA, and SQuAD generations using the paper's repetition-ratio proxy.

Dataset	Paper rho	Repro rho
HotpotQA	0.848	0.283
TriviaQA	0.820	0.593
SQuAD	0.856	0.386

All reproduced correlations stayed far below Table 3.

3. AUROC Does Not Match

HotpotQA detection landed far below Table 5.

0.663 FI AUROC	0.722 ENTROPY ONLY	0 REP>=.6 POSITIVES
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Entropy-only beat aggregate FI; top-quartile labels were needed for AUROC.

4. Scaling: 7B Reversal Fails

Lower mean entropy indicates stronger collapse. TinyLlama-Chat supports sub-3B instruction collapse, but Falcon-7B-Instruct still collapses more than Falcon-7B base.

The sweep is observational, matching the paper's caveat: architectures and alignment recipes vary, so the direct test is whether the reported direction appears under matched decoding.

Model	Type	Entropy
Falcon-RW-1B	Base	2.439
TinyLlama-1.1B-Chat	Instruct	1.118
OPT-1.3B	Base	2.492
Falcon-7B	Base	2.293
Falcon-7B-Instruct	Instruct	0.777

Each model used 20 HotpotQA prompts, matched sampling, and 48-token entropy traces. The reversal test is therefore direct: base and instruct Falcon-7B are compared within one model family.

Result interpretation: the instruct curve is non-monotonic only if it rises at

5. Hysteresis Helps, But Not 91-93%

The theta=0.50 / theta_low=0.40 deadband reduced alert flips, but the reduction was materially smaller than claimed.

Dataset	Static	Hyst.	Drop
HotpotQA	5.57	2.05	63%
TriviaQA	4.78	2.78	42%
SQuAD	6.03	1.77	71%

Mechanism: supported. Reported magnitude: not reproduced.

6. Stress Tests Split

NF4 quantization deepened entropy collapse, but evidence-position attention did not show the front 5-10x advantage.

-0.741 NF4 ENTROPY DELTA	1.15x FRONT/MID	0.68x FRONT/END
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Position	Attention
Front	0.01485
Middle	0.01293
End	0.02195

Overall Verdict

Claim	Status
Formula	Verified
Predictive rho	Falsified
Hotpot AUROC	Falsified
Scaling reversal	Falsified
Hysteresis magnitude	Falsified
Stress bundle	Mixed

The FI definition is reproducible; the paper's largest empirical effect sizes did not survive this scaled A100 audit.

Run Scope

- 350 real OPT-2.7B QA generations.
- Five-model scaling sweep, 20 prompts/model.
- Completed on A100-SXM4-80GB in about 12 minutes.
- All CSV/JSON artifacts pushed to the Hub dataset.

Artifact	Role
qa_summary.json	Claims 2, 3, 5
scaling_summary.csv	Claim 4